COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structures And Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	///26	**DATE SUBMITTED:**	//26
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HOA 11 Sorting Algorithms

1. PROCEDURE OUTPUT

ILO A: Measure the runtime of algorithms using theoretical analysis

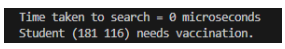
Assess if two arrays have any common values.

```
bool diff(int *x, int *y){
for(int i = 0; i < y.length; i++){
  if(search(x, y[i])!= - 1){
    return false;
     }
  }
 return true;
}
```

Performing worst case analysis:
Let m = the length of array x.
Let n = the length of array y.
The loop in diff repeats n times.
Each call to search requires m comparisons.
Given the above information, the total number of comparisons in the worst case is: **n^2**

Analysis:
Acala: As the size of array increases the number of operations also increases for example the size of arrow is 40 then the number of operations would be: 40^2 + 40 = 1640.

Esteta: As the size of the array increases the number of operations grows exponentially, The bigger the array size the longer the operation runtime.

Input Size (N)	Execution Speed	Screenshot	Observation(s)
1000	0 microseconds		Acala: As the sizes increase the execution

			speed also increases Esteta:The execution speed is faster with a smaller size
10000	1 microseconds	Time taken to search = 1 microseconds Student (181 116) needs vaccination.	Acala: As the sizes increase the execution speed also increases Esteta:The execution time of the search is faster than 100000 if they are looking for students deeper in the list otherwise its similar with the 100000
100000	2 microseconds	Time taken to search = 2 microseconds Student (181 116) needs vaccination.	Acala: As the sizes increase the execution speed also increases Esteta: The execution speed is the same with the 10,000 size if it is searching for the lower side of the list but is slower if they are searching deeper in the list.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO A: Measure the runtime of algorithms using theoretical analysis & ILO B: Analyze the results of runtime
performance by comparing experimental and theoretical analysis

For each of the problems below, provide the following:

- Theoretical Analysis
- Experimental Analysis
- Analysis and Comparison

Problem 1: Consider a program that returns true if the elements in an array are unique.

Algorithm Unique(A)
for i = 0 to n-1 do
for j = i+1 to n-1 do
if A[i] equals to A[j] then
return false
return true

	Acala	Esteta
- **Theoretical Analysis**	It uses for loop within a for loop so its a nested for loop therefore the worst time complexity would be O(n^2).	The algorithm should have a worst time complexity of n^2 because of the nested for loop.
- **Experimental Analysis**	`bool Unique(const std::vector<int>& A) { int n = A.size(); for (int i = 0; i < n - 1; i++) { for (int j = i + 1; j < n; j++) { if (A[i] == A[j]) return false; } } return true; }`	`bool Unique(const std::vector<int>& A) { int n = A.size(); for (int i = 0; i < n - 1; i++) { for (int j = i + 1; j < n; j++) { if (A[i] == A[j]) return false; } } return true; }`

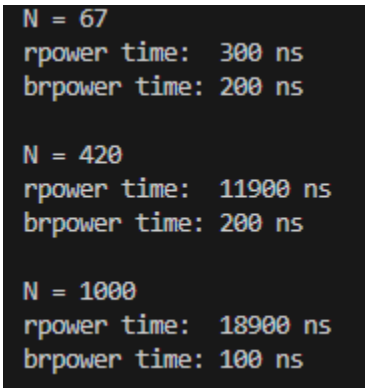
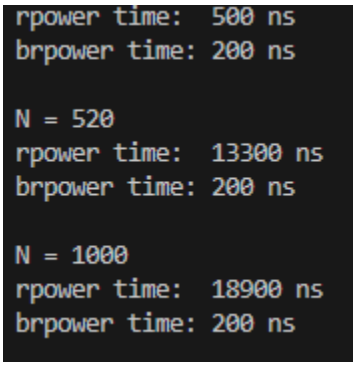
- **Analysis and Comparison**	`N = 1000 | Time: 1928 microseconds`
	`N = 2000 | Time: 7736 microseconds`
	`N = 4000 | Time: 31147 microseconds`
	The time search significantly increased as the data size increases. It justifies that the complexity is around n^2.
		`N = 1000 | Time: 1945 microseconds`
		`N = 2000 | Time: 7723 microseconds`
		`N = 4000 | Time: 31831 microseconds`
		The run time exponentially increases as the size gets bigger. This proves that the time complexity is N^2.

Problem 2: Consider another program that raises a number x to an arbitrary nonnegative integer, n.
Algorithm rpower(int x, int n):
1 if n == 0 return 1
2 else return x*rpower(x, n-1)
Algorithm brpower(int x, int n):
1 if n == 0 return 1
2 if n is odd then
3 y = brpower(x, (n-1)/2)
4 return x*y*y
5 if n is even then

6 y = brpower(x, n/2)
7 return y*y

	Acala	Esteta
- **Theoretical Analysis**	For rpower algorithm the time complexity would be O(n) while the brpower would be O(logn).	The first code block (rpower) has a worst time complexity of O(N) while the second code block(brpower) has a worst time complexity of O(log n)

- **Experimental Analysis**	`long long rpower(long long x, int n) {`
	`  if (n == 0) return 1;`
	`  return x * rpower(x, n - 1);`
	`}`
	
	`long long brpower(long long x, int n) {`
	`  if (n == 0) return 1;`
	`  if (n % 2 != 0) {`
	`// odd`
	`  long long y = brpower(x, (n - 1) / 2);`
	`  return x * y * y;`
	`  } else { // even`
	`  long long y = brpower(x, n / 2);`
	`  return y * y;`
	`  }`
	`}`
		`long long rpower(long long x, int n) {`
		`  if (n == 0) return 1;`
		`  return x * rpower(x, n - 1);`
		`}`
		
		`long long brpower(long long x, int n) {`
		`  if (n == 0) return 1;`
		`  if (n % 2 != 0) {`
		`// odd`
		`  long long y = brpower(x, (n - 1) / 2);`
		`  return x * y * y;`
		`  } else { // even`
		`  long long y = brpower(x, n / 2);`
		`  return y * y;`
		`  }`
		`}`

- **Analysis and Comparison**	 As the data size increases the rpower time also increases since the time complexity is O(n) but the brpower stays relatively low since its O(log n) therefore all through out the data size i tested the brpower time almost stays the same.	 The time complexity of rpower and brpower vastly differ and grows even larger as the size grows bigger Because rpower has a run time of O(N) while brpower has a run time of O(log n).

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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